

Design and Development of a USB-Powered Smart Anti-Theft System for Personal Devices Using Arduino with Motion Detection and Alert

Abstract

This project presents the design and development of a USB-powered smart anti-theft system for personal devices using an Arduino microcontroller with motion detection and real-time alert functionality. The system is designed to protect valuable personal devices such as laptops, bags, and portable electronics from unauthorized movement or theft. Powered through a standard 5 V USB source, the system is compact, portable, and energy-efficient. A motion sensor continuously monitors the protected object, and when unexpected movement is detected, the Arduino immediately activates an audible buzzer and LED warning indicator to alert nearby users. The system can also be configured to send notifications through a wireless communication module as a future enhancement. Its low-cost design, simple implementation, and reliable operation make it suitable for students, offices, libraries, and public spaces. The proposed solution demonstrates an effective and affordable approach to enhancing personal device security using embedded system technology.